Natural and Robust Humanoid Locomotion using Reinforcement Learning

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***Abstract*—This work develops reinforcement learning-based locomotion for the Booster T1, a humanoid robot whose long, heavy arms and dexterous hands give it an asymmetric morphology that the official controller does not account for, resulting in a short, stompy gait with the arms held retracted. A legged controller is first trained and deployed on the real robot, and then extended to the whole body, with the arms swinging in anti-phase with the legs. Compared with the official controller, the deployed whole-body policy reduces mean mechanical power by 41% and torque-rate RMS by 29.5%, and a second formulation that minimizes the cost of transport shows that a restrained arm swing is the economical solution for this morphology. Walking is then enhanced with reference motions retargeted from an established motion library and our own recordings, organized by speed so that the gait is more pronounced at high velocities and restrained as it slows down. This policy walks with a less crouched posture and converges in roughly 65% fewer training steps. Finally, the framework is extended to perceptive stair traversal. A policy that observes a heightmap of the terrain ahead learns ascent and descent as structured sub-movements and, in simulation, climbs steps of up to 25 cm. It is blended with the walking policy so that neither behavior degrades and transitions between them remain smooth. Every stage is replicated on six further humanoid robots. In summary, this work contributes: (i) a whole-body locomotion controller for an asymmetric humanoid, deployed on the real robot and compared at the joint level with its official controller; (ii) a cross-morphology study of energy-efficient walking at equal Froude number; (iii) velocity-indexed reference motions, retargeted for non-human proportions, that make the gait change character with speed; and (iv) a perceptive climbing policy built from explicitly rewarded human sub-movements and blended with the walking policy.**

***Index Terms*—Humanoid Locomotion, Reinforcement Learning, Reference Motion, Perceptive Stair Traversal, Policy Blending, Sim-to-Real Transfer.**

I. Introduction

BIPEDAL locomotion remains one of the central challenges of humanoid robotics. A humanoid is an inherently unstable system with numerous coupled joints, and its controller must stay reliable under any disturbances, measurement noise, and the latency of real hardware. Model-based approaches such as Model Predictive Control address this through an explicit model of the dynamics, but the cost of solving it online, and the sensitivity to modeling error, force strong simplifications, such as reduced upper-body dynamics or a single fixed gait [1]. Deep reinforcement learning (DRL) instead learns the controller offline in simulation, across thousands of parallel environments, and has become the dominant approach for deployable humanoid locomotion [2].

Recent work has also shifted the goal from merely stable walking towards natural, life-like motion [3], [4]. Abrupt robotic movements are unpredictable around people, and tend to consume more energy than smooth, controlled ones. A less explored front is the locomotion of humanoids exhibiting asymmetric morphologies, whose arms are disproportionate to their legs. This work sits at the intersection of the two: its core is natural and robust movement, on a robot whose morphology makes both harder to obtain.

The robot used is the Booster T1 in its manipulation variant, with three additional degrees of freedom per arm and dexterous hands. The added mass and length of the arms, which hang down to the knees, change the dynamics of the robot considerably. The official controller supports only legged locomotion, keeps the arms permanently retracted in front of the body to avoid self-collision, and walks with very short, stompy steps. It is also proprietary and distributed only in pre-compiled format.

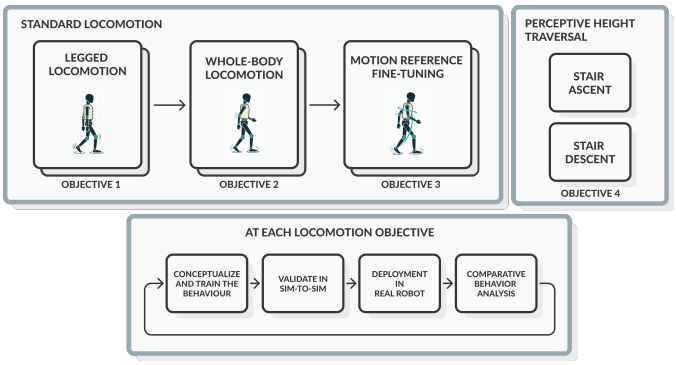


Fig. 1. Structure of the Work. Each locomotion objective follows the same cycle of training, sim-to-sim validation, real robot deployment and comparative analysis.

The work is organized as a progression of four objectives (Fig. 1). A legged controller with a fixed upper body is first deployed and compared directly with the original Booster controller. It is then extended to whole-body locomotion, under an unrestrained objective in which the arms move freely, coordinated only with the legs, and under an energy efficiency objective in which arm motion is left to emerge from minimizing the cost of transport. Third, the walking is enhanced with reference motions organized by walking speed. Finally, the controller is extended to perceptive stair ascent and descent, learned as structured sub-movements and blended with the reference motion walking. Every approach is also trained on six further humanoids, to examine how each behavior holds across stature and morphology. In summary, this work makes the following contributions:

- a legged and a whole-body locomotion controller for an asymmetric humanoid, deployed on the real T1 and compared at the joint level with the official controller, with lower mechanical power and smoother motion;
- a study of energy-efficient whole-body walking across five humanoids at equal Froude number, showing that the economical arm swing follows each robot's morphology;
- an adaptation of interaction-mesh retargeting to robots of non-human proportions, and a velocity-indexed library of human gait cycles that makes the gait more pronounced at high speed and restrained at low speed;

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- a perceptive stair ascent and descent policy whose rewards follow the sub-movements a person performs, trained off-policy on a heightmap with realistic imperfections, and blended at deployment with the walking policy;
- a common training and validation pipeline for every objective, transferred to six further humanoid robots.

Videos of all the developed work are available in a single video playlist. Part of this work was presented at the Portugal–U.S. Partnerships Summit, in October 2026, as part of a joint research project between ISR-Lisbon and the University of Texas at Austin.

II. Background and Related Work

A. Off-Policy Learning for Humanoid Locomotion

Most humanoid locomotion controllers are trained with PPO [2], whose clipped objective yields stable updates. Off-policy algorithms such as TD3 [5] instead reuse every transition through a replay buffer, which makes them considerably more sample efficient. This property matters here. The T1's asymmetric weight terminates episodes earlier and more often than on a proportionate robot, the whole-body and stair objectives enlarge the learning problem, and stair terrain multiplies the diversity of training conditions. FastTD3 [6] recently made TD3 practical with massively parallel simulation, large batch updates and a distributional critic. It is adopted here as the learning algorithm, with an asymmetric actor-critic structure [7] in which the critic receives privileged simulator information.

B. Natural Movement through Reference Motions

Handcrafted rewards decompose locomotion into task objectives and reach a plateau in movement fluidity, as they are limited to what the designer explicitly encodes. Imitation learning takes the opposite route, learning the style of human data directly through motion tracking or adversarial motion priors [8]. Pure imitation, however, tends to reproduce a fixed choreography, tied to the recorded clips and less responsive to commands and disturbances. A middle ground uses the reference as a tool within an established reinforcement learning reward structure, first demonstrated on simulated characters [9], so the reference shapes how the robot moves while the task rewards keep it robust and responsive to commands. References come from motion-capture datasets such as LAFAN1 [10], transferred onto the robot through retargeting algorithms. Interaction-mesh formulations such as OmniRetarget [11] preserve the spatial relations between body parts by matching Laplacian coordinates of a volumetric mesh over the keypoints [12], and enforce joint limits, collisions and foot contact as hard constraints. These methods have mostly been tuned for humanoids of near-human proportions. The datasets were also recorded for other purposes, such as game development, with walking interleaved with unrelated movements, and a single extracted gait cycle fixes one walking style at one speed, whereas human gait changes character with speed.

This work departs from existing approaches in four respects. The retargeting is adapted to robots whose proportions differ from the human's. Individual gait cycles from an established dataset and dedicated recordings at multiple speeds are organized into a library indexed by walking speed, so the gait changes character with the commanded velocity. And the reference covers the whole body, including the arms, whose swing matters most for robots with long and heavy upper limbs. Finally, the resulting policy is deployed on the real robot, where the reference-guided gait transfers faithfully from simulation.

C. Energy Efficiency and Emergent Gaits

The cost of transport (CoT), the energy spent per unit weight per unit distance, $\mathrm{CoT} = P/(mgv)$, is the standard dimensionless measure of locomotion efficiency [13]. Unlike a penalty on power, which can be minimized by not moving, it prices energy per distance. Minimizing energy alone has been shown to be enough for natural gaits to emerge in legged robots [14], but these studies treat the legs only. In humans, the arm swing lowers the metabolic cost of walking by cancelling the angular momentum of the legs [15]. Whether such a swing emerges in a humanoid from the energy objective alone, and how it depends on arm mass and length, has not been examined. This work studies this question across humanoids of different morphology, comparing robots of different size at equal Froude number, $\mathrm{Fr} = v^2/(gL)$, at which geometrically similar walkers use dynamically similar gaits [16].

D. Perceptive Stair Traversal

Blind policies infer the terrain through contact and have climbed stairs on humanoids [17], but they only detect a step once a foot strikes it. They are therefore conditioned to recover after the toe hits the step rather than to plan the placement, a reflex that becomes more dangerous as the step grows taller relative to the robot. Perception of the terrain ahead substantially increases stair success [18], either from depth images [19], [20], which are costly to render across parallel environments, or from a heightmap [21], [22]. Foot placement is especially critical for humanoids, which stand on two feet of finite length [23]. A regulation step of 18 cm reaches about a third of the T1's leg length, and its blocky feet leave little margin on each tread, so each step must be planned.

This work addresses these requirements in three ways. Stair traversal is decomposed into the sub-movements a person performs, each rewarded explicitly, which gives the most controlled and precise placement at every stage. Instead of one network for both behaviors [24], a dedicated climbing policy is blended at deployment with the reference motion walking policy, so each expert can specialize fully. And the climber is trained off-policy on a heightmap whose imperfections approximate those of the real camera, which keeps the cost of training affordable and allows each sub-movement to be trained until it is accurate.

III. Methodology

A. Simulation Pipeline and Robot Models

Training and validation environments were built for both MuJoCo and NVIDIA Isaac, with the same observation processing and action computation, so that every policy can be cross-validated on a second physics engine. MuJoCo, lightweight and real-time, is used for fast iteration and for joystick-controlled validation under the same conditions as real deployment. Isaac provides terrain generation and the validation of hundreds of robots at once. A third simulator, Webots, provided by Booster in a pre-compiled format, implements the robot's SDK and is used to test the deployment script, the exact code run on the real robot, before it reaches the hardware. Its scene was extended with staircases and the conversion of the camera point cloud into a heightmap.

The official models describe only the simplified 24-DoF T1, so the manipulation arms were extracted from the robot's USD file and the XML model rewritten around them. Six further robots were modeled to the same completeness (Fig. 2): the Unitree G1, Booster K1, Fourier GR3, Agibot A3, PAL Kangaroo and Deep Robotics DR02. Their joints were mapped to a common layout and their actuation normalized to mass and leg length, so that gait differences reflect morphology rather than how the robot's model was set up.

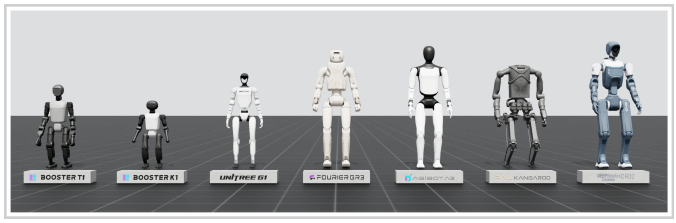


Fig. 2. The Seven Humanoid Robots Studied, Standing at Rest. The T1 is the second shortest, yet its arms reach below its knees.

B. Legged Locomotion

The first policy controls only the 12 leg joints, with the upper body fixed. It observes the velocity command $\mathbf{c} = (v_x, v_y, \omega_z)$, a gait phase $(\cos 2\pi ft, \sin 2\pi ft)$, the gravity vector and angular velocity from the torso IMU, the joint positions and velocities, and the previous action, and outputs joint position targets at 50 Hz. The critic additionally receives privileged quantities, such as base linear velocity, base height, push forces and foot velocities. The reward follows standard practice for humanoid locomotion: velocity tracking and survival, penalties on torso orientation, base height, torque, joint velocity and acceleration, joint limits and collisions, and phase-based rewards that lift the left and right feet in swing windows centered at phases 0.25 and 0.75. Training uses TD3 with parallel environments [6], two critics and a delayed actor updated several times per global step from a replay buffer that samples evenly across environments. Two arm configurations were trained: retracted, matching the official controller, and lowered along the torso.

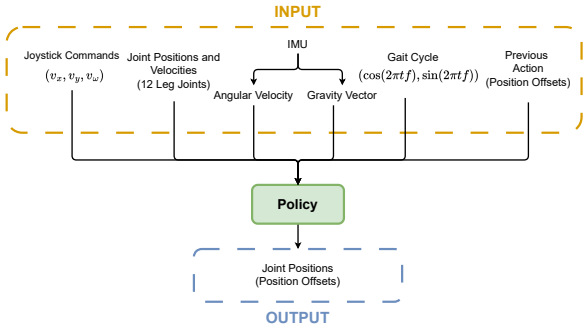


Fig. 3. Structure of the Legged Policy. The actor receives the velocity command, the gait phase, the IMU readings, the leg joint states and its previous action, and outputs leg joint position targets that are tracked by the robot's joint controllers.

C. Whole-Body Locomotion

Adding the shoulder and elbow joints to the action space with the baseline reward produced erratic arm motion. Three deficiencies motivated three changes. Arm-to-leg contacts are penalized with one of the largest weights in the reward, as a near-hard constraint. Admissible windows on shoulder roll and pitch prevent the policy from raising and holding the arms to redistribute its mass, a local optimum far from human gait. And contralateral coordination is made a precondition of the foot-swing reward, which is granted during the swing of leg a only when the opposite shoulder pitches in the same direction as the swinging hip,

$$1^{(a)}_{\text{limbs follow}} = 1_{\dot{q}^{\text{sh}}_{\bar{a}}\, \dot{q}^{\text{hip}}_{a} > 0}, \quad a \in \{\text{L}, \text{R}\}. \tag{1}$$

This prescribes the direction of the swing but not its shape, which is left to the reference motions of the next objective.

Under the energy efficiency objective, arm motion is not rewarded at all, and the CoT is the main optimization target, computed per step with positive mechanical power,

$$\text{CoT} = \min\left(\frac{\sum_j \max(\tau_j \dot{q}_j, 0)}{m\, g\, \max(|v_x|, v_{\min})},\ c_{\max} \right), \tag{2}$$

with $v_{\min} = 0.15$ m/s to avoid the singularity near standstill and $c_{\max} = 3$ to keep a stumble from dominating the reward. A power penalty was tried first, but at a third of the objective the T1 walked at only 0.08–0.10 m/s for commands of 0.30–0.50 m/s. The same environment was trained on the other robots, which are compared at the speed matching the Froude number of the T1 at 0.46 m/s (Fr ≈ 0.04). The swing is quantified by the mass-weighted measure $(m_{\text{arm}}/m_{\text{body}})(\Delta x_{\text{arm}}/L_{\text{leg}})$, which combines the arm's share of the body mass with how far its center of mass travels relative to leg length.

D. Reference Motion Fine-Tuning

Retargeting: OmniRetarget scales the human targets by a constant fitted to the Unitree G1. The T1 is 1.20 m tall, about 29% shorter than the average human, and the unmodified algorithm over-scales its targets by about 10%. The robot then walks on its toes with its feet never reaching the floor, and the arm joints saturate at their limits, while the solver still converges at an unremarkable cost. The pipeline was modified at three levels (Fig. 4). The scale is set relative to the robot's height and calibrated so the soles touch the floor. The mesh nodes are selected by how observable they are through the robot's Jacobian, rather than by anatomical name, since a same-named link may sit elsewhere, such as the T1's ankle frame well above its sole. And a soft term draws contacting soles flat onto the ground, while a hard constraint keeps a minimum lateral separation between the feet,

$$\begin{aligned} q_t^\star = \arg\min_{q_t}\ & E_{\text{mesh}} + \|q_t - q_{t-1}\|^2_Q + E_{\text{sole}} \\ \text{s.t.}\ & n(q_t)^\top (f_l(q_t) - f_r(q_t)) \geq d_{\min}, \end{aligned} \tag{3}$$

$$\begin{aligned} E_{\text{mesh}} &= \textstyle\sum_i \|L(p^{\text{source}}_{t,i}) - L(p^{\text{target}}_{t,i}(q_t))\|^2, \\ E_{\text{sole}} &= w_g \textstyle\sum_{i \in \mathcal{C}_t} s_{t,i} \left(e_z^\top f_i(q_t) - \bar{z}_{t,i}\right)^2, \end{aligned} \tag{4}$$

in addition to the original constraints, where p^{source} and p^{target} are the human and robot keypoints, f_i the sole keypoints, $\mathcal{C}_t$ those in contact, $\bar{z}_{t,i}$ a floor target approached by at most δ per frame, and $s_{t,i}$ a fade before lift-off that lets the heel roll off naturally.

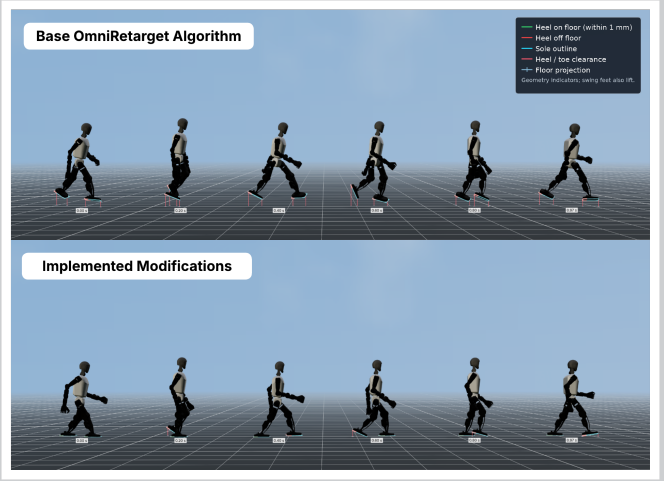


Fig. 4. The Same Walking Clip Retargeted onto the T1 with the Base OmniRetarget Formulation (Top) and with the Implemented Modifications (Bottom) (Video).

Velocity-indexed library: Gait cycles are extracted from the retargeted robot motion between consecutive right-foot touchdowns, and scored as looping references by

$$S_{ab} = \|\mathbf{x}_a - \mathbf{x}_b\| + 2\,\frac{|T_{ab} - \tilde{T}|}{\tilde{T}} + 0.02\,|\Delta\psi_{ab}| + 5 \cdot 1_{n_L \neq 1}, \tag{5}$$

which favors cycles that close on themselves, have a typical duration, do not turn, and contain exactly one left touchdown. Selected cycles are normalized to a common 49 frames, which fixes a single gait frequency of about 1.02 Hz, and phase-aligned so their swing apexes match the environment's foot-swing windows. The library holds 26 cycles spanning 0.23–1.28 m/s, 18 from our own OptiTrack recordings at lower speeds and 8 from LAFAN1 at higher speeds. At each step, the policy tracks the cycle whose captured speed is closest to the commanded speed. Because all cycles share one phase clock, a change in the command changes the style of the reference without disturbing its timing.

Imitation: Arms and legs are rewarded separately, $r_G = 1_{\|\mathbf{c}\|>0.1} \exp(-\frac{1}{\sigma_G} \sum_{j \in G} w_j (q_j - \hat{q}_j)^2)$, with weights 2.0 and 1.5, so the error of one group does not saturate the other's signal. Per-joint weights emphasize the joints that carry the character of the gait, such as hip, knee and shoulder pitch, over the lateral balance joints. The indicator disables imitation when standing still, where a position-only term holds the home pose. The arm windows of the whole-body objective were removed, and the base-height penalty relaxed, since the reference now supplies those behaviors.

E. Perceptive Stair Traversal

The walking policy becomes unstable on steps above about 4 cm. Placing stairs in the environment and relying on velocity tracking only teaches the robot to stop at the first step, while rewarding altitude with proprioception alone reproduces the blind stumble-and-recover strategy. The target is the 18 cm step of building regulations, which for the T1 is proportionally like a person climbing 36 cm steps for an entire flight.

Terrain and curriculum: Stairs are built as an inverted pyramid of 12 rings around a central platform, so walking outward ascends and inward descends. Each environment samples a first step of 2–4 cm and a final step of 8–18 cm, reached linearly over eight rings, with tread depths of 0.4–0.8 m, and 20% of environments are anchored at 18 cm. A later ascent-only stage extends 30% of flights to 18–30 cm steps to find each robot's limit. Most episodes start on the stairs, so the policy meets steps often rather than learning to stop at them.

Perception: The climber observes a 5×5 heightmap spanning 0.1–0.7 m ahead and ± 0.3 m to the sides, referenced to the terrain under the base so it is zero on flat ground. During training, domain randomization introduces imperfections into the observed heightmap like those of the real head camera: partially observed rows, 30 Hz frames arriving late, feed drops, noise, outliers, rows under the feet filled from memory, and shifts from localization error. The observed heights differ from the exact ones by a median of 0.8 cm and a 95th percentile of 2.7 cm. The exact heights are kept for rewards and terminations. The climber is warm-started from the walker, with the 25 new input weights initialized to zero, so it initially walks exactly like the walker.

Step engagement: The walking terms that oppose climbing, such as reference imitation, the swing clock and smoothness penalties, are not removed but released by a continuous engagement signal $w \in [0, 1]$ computed from the heightmap. For ascent, it ramps with the highest point h_{near} of the two nearest heightmap rows,

$$w_{\text{asc}} = \text{clip}\left(\frac{h_{\text{near}} - 0.005}{0.018 - 0.005},\, 0,\, 1\right), \tag{6}$$

and is held at 1 while a foot is at the step or still in the air on a tall step. Descent engagement is computed in the same way from the drop ahead, and $w = \max(w_{\text{asc}}, w_{\text{desc}})$. The ramp avoids a discontinuous reward as a function of a continuous observation, which a value function fits poorly. Imitation is released by up to 95% on tall steps, smoothness and energy penalties by 60%, the gait clock slows to between 75% and 50% of its cadence, and the hip and knee flexion range is extended by up to 1.35 times within the joint and torque limits. On flat ground, $w = 0$ and the walking objective is untouched.

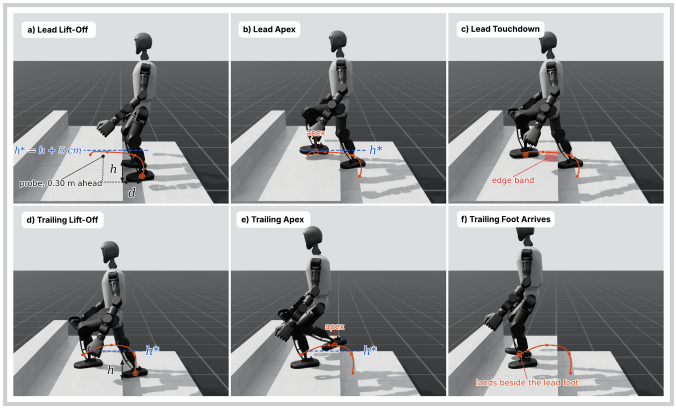


Fig. 5. A Single Ascent Step onto a 17.9 cm Step from a Rollout of the Climbing Policy. Top: lead foot, from lift-off and apex to touchdown on the upper tread. Bottom: trailing foot, meeting the lead foot on the same tread.

Ascent: The rewards follow the sub-movements of a person climbing step-to-step (Fig. 5): approaching the step, placing the stance foot at a distance that leaves room for the swing, lifting the swing foot above the step, placing it fully on the tread, bringing the trailing foot beside it, and settling before the next step. Most are event rewards paid at touchdown. The foothold reward is a Gaussian on the distance d from the sole to the step, around a target that moves back as the step grows. The required clearance h^* is the step height plus 5 cm, set at lift-off from the terrain ahead, and the swing is penalized when it falls below a smooth rise schedule,

$$c_{\text{clear}} = \sum_{\text{airborne } f} \max\Big(0,\, h_f^* \mathcal{S}\big(\tfrac{t_{\text{air},f}}{T_{\text{rise}}}\big) - \big(z_f^{\text{peak}} - z_f^{\text{liftoff}}\big)\Big), \tag{7}$$

with $\mathcal{S}(x) = x^2(3 - 2x)$, which shapes the swing towards the real obstacle without rewarding an explosive kick. A tempo reward matches the air time to inverse-dynamics estimates of the achievable swing, rising is paid only for new maxima of base height, and touchdowns on the edge, scrapes on the step face and lifts without a settled double stance are penalized. A stall timer that only real upward progress can reset makes waiting in front of a tall step strictly worse than attempting it.

Descent: The policy infers ascent or descent from the sign of the heightmap. Descent rewards new supported minima of foot height, paid once per level, penalize the downward speed of the base and of the landing foot above 0.35 m/s, so the robot cannot earn progress by falling, and reward landing on a plateau over the distance d past the crossed edge,

$$r_{\text{land}}(d) = \begin{cases} 0, & d < 0.12, \\ \text{ramp } 0 \to 1, & 0.12 \le d < 0.16, \\ 1, & 0.16 \le d \le 0.24, \\ \text{decay } 1 \to 0, & 0.24 < d \le 0.35, \end{cases} \tag{8}$$

since ascent needs a precise stance distance but descent only needs the foot to land safely, clear of both edges. Below 12 cm the heel

still overhangs the lip, beyond 0.35 m the step becomes a lunge, and because the band is measured from the crossed edge it does not depend on tread depth (Fig. 6).

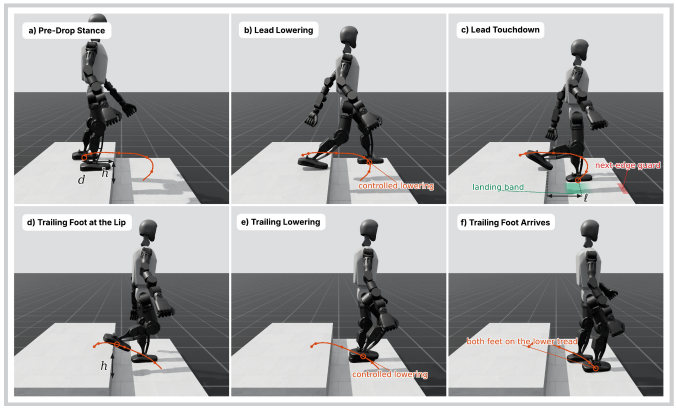


Fig. 6. A Single Descent Step Down a 17.9 cm Drop from a Rollout of the Climbing Policy. The lead foot is lowered past the lip and lands within the rewarded band 0.16–0.24 m beyond it, clear of the next edge, before the trailing foot joins it on the lower tread.

Touching down within 4 cm of the next edge is penalized, and the feet are guided towards a prepared stance before each drop.

Blending: The climbing policy still imitates the reference on flat ground, so that it walks soundly if the blend misfires, but its walking is never as clean as that of the reference motion policy. The two are therefore combined at deployment by a gate on the heightmap,

$$\begin{aligned} \mathbf{a} &= \beta\, \mathbf{a}_{\text{walk}} + (1-\beta)\, \mathbf{a}_{\text{climb}}, \\ \beta^{\text{inst}} &= 1 - \text{clip}\Big(\tfrac{\max|\text{hm}| - 0.02}{0.05 - 0.02}, 0, 1\Big), \end{aligned} \tag{9}$$

so both steps and drops engage the climber. The gate follows β^{inst} downwards immediately, handing control to the climber as soon as a step comes into view, but returns to the walker by at most $1/40$ per step, which keeps control from bouncing between policies when the heightmap flickers. Since both policies share the action space, home pose and phase clock, and the climber was warm-started from the walker, their actions are compatible near the switch. For the DR02, an alternating step-over-step stride was added on low, shallow stairs where reachability conditions on tread depth, step height and leg span hold, with a fallback to the step-to gait elsewhere.

Camera pipeline: At deployment, the head is pitched down and each depth frame is registered with the camera pose. Points on the robot's limbs and on vertical step faces are removed, the rest accumulated in a world-frame elevation map of 4 cm cells that leaves ambiguous cells empty rather than averaging across an edge, and the 25 heights are sampled from it as in training. This pipeline is too costly to run per agent during training, so it is validated with a simulated camera, through the deployment script in Webots, and with point clouds recorded by the real head camera and replayed in simulation.

IV. Experiments and Results

A. Experimental Setup

All policies were trained on a single RTX 5080 GPU. On the real T1, they run at 50 Hz through the Booster SDK, publishing joint position targets with the same stiffness and damping as in training, and are teleoperated with a joystick. Joint positions, velocities and torques are recorded from the robot's state topic. Gaits are compared through joint ranges, the deviation between time-normalized strides relative to each joint's range, mechanical power $\overline{|\tau\dot{q}|}$ and positive power, torque rate and jerk, and the left–right similarity of joint waveforms. The energy study compares robots at Froude-matched speeds over up to six seeds, measuring 11 s of straight walking after 7 s of settling. Stair success is measured over 400 randomized-start episodes per condition, on a uniform 12-step flight with a 0.5 m/s forward command and an 80 s limit, at three tread depths and in both MuJoCo and Isaac Sim. An episode succeeds only if the full flight is climbed without a foot misplacement or loss of balance.

B. Legged Locomotion

Training took about 12 hours on a single RTX 5080 GPU, and the reward plateaued after about 8% of the run. Checkpoints were saved at fixed intervals and the deployed policy was selected in joystick-controlled MuJoCo validation as one showing no unnatural motion and never losing balance under manual stress tests. The policies were also run in Isaac Sim and Webots, with MuJoCo remaining the closest approximation to the real robot. Both arm configurations transferred directly to the real robot with behavior very similar to simulation. Against the official controller (Fig. 7), the policy moves with a range 2.1 times larger at the hip, 1.4 times at the knee and 1.8 times at the ankle, while keeping every pitch joint at least 15° from its limits. Its strides are also more consistent, with the deviation between normalized strides falling from 12% to 7.5% at the hip and from 21% to 12% at the ankle. It walks at 1 Hz against the official 1.9 Hz, with fewer and longer steps. The hip pitch leads the movement and spans the widest range, producing about two thirds of the ankle's horizontal displacement relative to the hip, while the knee and ankle mainly keep the foot clear of the ground during the swing and level it for touchdown. In simulation and on the robot the hip ranges match within 2°, with the ankle, most affected by ground contact, showing the largest gap. Lowering the arms, even while fixed, reduced the range of the leg pitch joints by about 13%, and the knee's by up to 23%, easing the load on the legs. The lowered configuration also looked more stable and poised under abrupt changes of command, such as fully stopping while walking backwards.

C. Whole-Body Locomotion

The whole-body policy swings its arms in anti-phase with the legs, conservatively, and transfers onto the real robot with a cycle-averaged RMSE of 1.8° against simulation, leg correlations of 0.99 and arm correlations of 0.95–0.99. The mean posture of every joint agrees within 1.5°, and the hip and knee pitch deviate by only 6% of their range. The anti-phase swing is clearly present in both simulators, but much fainter than in average human walking. Freeing the arms improves stability during rapid stops, as they counterbalance the center of mass, and changes the lower body as well, with reduced knee flexion.

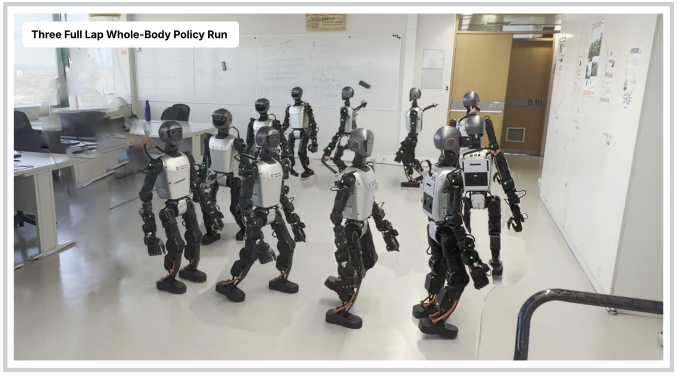


Fig. 8. Three-Lap Robustness Trial of the Deployed Whole-Body Policy (Video).

This gives smoother and more controlled stops, which matter for when small adjustments are required, such as before any manipulation tasks. Some left-right asymmetry appears in the active arm joints, since symmetry is not enforced during training, and it is one of the reasons for the reference motions of the next objective, whose symmetric cycles enforce it.

A three-lap trial (Fig. 8) tested whether the arm motion raises the overall demand. Compared with the original Booster controller on the same route, it reduced mean absolute mechanical power by 41.2%, positive power by 40.3% and torque RMS by 16.4%, and its motion was smoother, with 29.5% lower torque-rate RMS and 80.7% lower jerk (Table I). The arm swing does add upper-body demand, with recurring shoulder-pitch torque peaks, but this is small relative to the hip and knee reductions, and together with a higher ankle demand it is outweighed by the lower-body savings, so the arms are an asset rather than a burden for this morphology.

TABLE I: Aggregated Metrics across All Joints over the Three-Lap Trial.

Metric	Booster	Whole-Body	Red. [%]
Mean $\lvert\tau\rvert$ [Nm]	106.3	82.4	22.5
RMS τ [Nm]	132.7	111.0	16.4
$\overline{\lvert\tau\dot{q}\rvert}$ [W]	91.3	53.7	41.2
Positive power [W]	58.8	35.1	40.3
RMS $\dot{\tau}$ [10^3 Nm/s]	10.2	7.2	29.5
RMS jerk [10^3 rad/s^3]	806.3	156.0	80.7

The whole-body policy also shows smaller torque fluctuations at the hip and knee, in particular without the repeated negative knee-torque peaks of the official controller. A smaller drop in battery charge was also observed over the route, consistent with the lower mechanical power.

Under the energy efficiency objective, the T1 converged to the same conservative arm swing, even though the whole-body policy had been free to adopt larger arm movements, which indicates that this swing is the economical solution for its morphology. The energy efficiency policy, however, adopts a more flexed leg posture, lowering the arms closer to the ground. This limits their influence on balance while preserving the swing, so both policies reach similar arm behavior through different postures. Across the other robots (Table II, Fig. 9), the economical swing follows each morphology rather than a universal pattern.

TABLE II: Arm Morphology and Swing of the Cost-Focused Policies at Froude-Matched Speed.

Metric	T1	G1	K1	GR3	A3
Leg length [m]	0.52	0.65	0.45	0.91	0.88
Arm reach / leg	1.13	0.82	0.91	0.69	0.82
Speed [m/s]	0.46	0.53	0.42	0.60	0.58
Shoulder pitch range [°]	16.4	24.0	41.0	11.0	21.3
Hand travel [cm]	12.4	15.6	12.1	10.6	15.7
Mass-weighted swing [10^{-3}]	12.9	14.4	12.6	5.0	7.2

The T1's arms reach 1.13 times its leg length, against about 0.83 for a human [25], and each carries 10% of its mass. The robots closest to human proportions, the K1, G1 and A3, swing their arms visibly more and stay upright. The GR3, with short and bulky arms, barely moves them. The T1 rotates its shoulder only 40% as much as the K1, yet its long arms act as levers, and its mass-weighted swing is nearly identical.

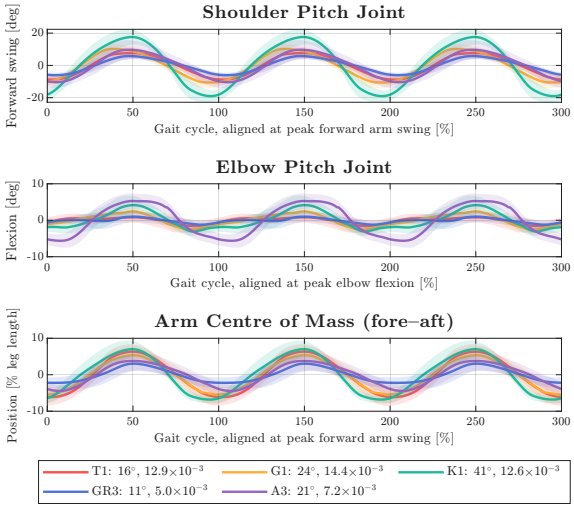
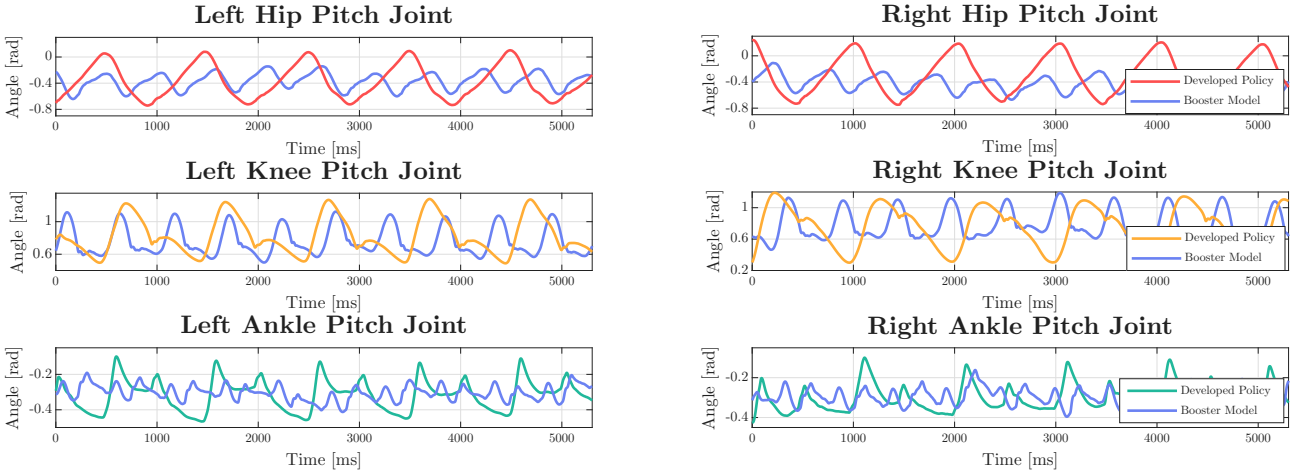


Fig. 9. Arm Swing of the Cost-Focused Policies over One Gait Cycle at Froude-Matched Speed. From top to bottom: shoulder pitch, elbow pitch and fore-aft arm center-of-mass position (% of leg length), mean posture removed, mean ± 1 SD.

Efficiency alone, therefore, does not lead the T1 to a human-like gait, and the desired characteristics have to be specified explicitly. In



(a) Left leg. (b) Right leg.

Fig. 7. Leg Pitch Joint Positions over a Forward Walk on the Real Robot: Developed Legged Policy against the Original Booster Controller. The policy moves through a wider range, with longer and more consistent strides. (Video)

the same rollouts, the K1, G1 and A3 kept upright postures, whereas the T1 adopted its flexed stance.

Economy and human-likeness do not coincide for every morphology. For a robot whose upper body is far from human proportions, such as the T1's, the most economical walk is neither upright nor pronounced in its arm swing. The robots with more human-like arms move closer to a human-like swing under the same objective, so the T1's restraint comes from its morphology rather than from a shortcoming of the cost objective.

D. Reference Motion Based Locomotion

Our own recordings, retargeted with the modified pipeline, yield robot-specific references for every robot (Fig. 11). Although the robots reproduce the same captured movement, differences in stature, limb proportions and degrees of freedom lead to different joint configurations. The smaller T1 and K1 cannot reproduce some movements of the 1.90 m subject exactly, while robots closer to the subject in proportions follow the motion more faithfully, but the coordination between the stepping legs and the swinging arms is preserved in all of them. Each reference describes a complete walking cycle at a given speed, so the desired gait is depicted directly at every speed instead of being encoded through separate handcrafted rewards, and together the references span the range of velocity commands used in training. The retargeted motions are purely kinematic, so the policy does not replay them, but approximates each reference while still walking, balancing and following its commands. The added retargeting parameters were tuned for each robot, and the resulting policies were assessed with the same validation procedure as the previous objectives.

With reference guidance, the T1's arm swing gains visible elbow flexion faithful to the recordings, and the velocity conditioning holds (Fig. 12).

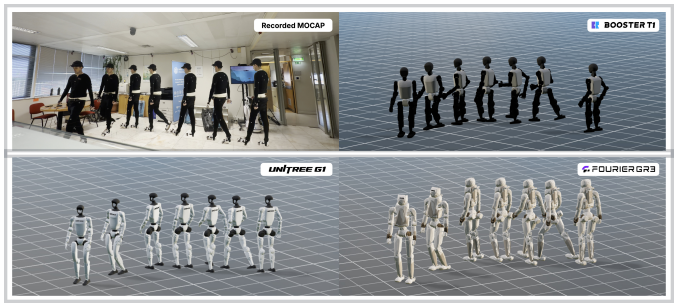


Fig. 11. Excerpt of Our Recorded Motion Capture and its Retargeting to the T1, G1 and GR3 (Video).

At maximum speed, the alternating arm motion is accompanied by elbow flexion, giving the swing a more articulated form than moving a nearly fixed arm forwards and backwards. At low speed the arms stay close to the trunk, much as in the whole-body policy, and both stride and swing grow with the commanded speed. The pronounced, articulated swing is therefore specific to fast walking rather than applied uniformly at every speed.

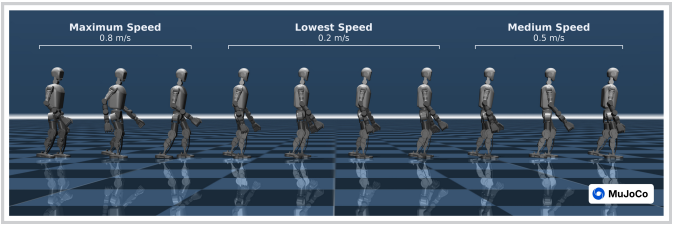


Fig. 12. Velocity-Conditioned Behavior of the Reference Motion Policy in MuJoCo. At low commanded speed the arms stay close to the trunk, and both stride and arm swing grow as the speed increases.

The references also accelerated learning. Without them, the reward first stalls well below its final value while the agent discovers a walking gait from task and regularization rewards alone. On the T1, training stabilized in roughly 65% fewer steps, as the walking pattern is supplied from the start instead of being discovered from task rewards. The effect was milder on the G1 and absent on the K1, which already finds a stable gait quickly (Fig. 10), which suggests the reference helps most where walking is hardest to discover.

On the other robots, all policies adopted a similar alternating arm swing, in contrast with the energy objective, where the arm motion differed between robots. The G1 and DR02 use their waist pitch joint to reproduce the subject's forward lean, while the bulky GR3 converged to an upright posture, as leaning proved less stable, so the reference is followed only as far as it remains compatible with stable walking.

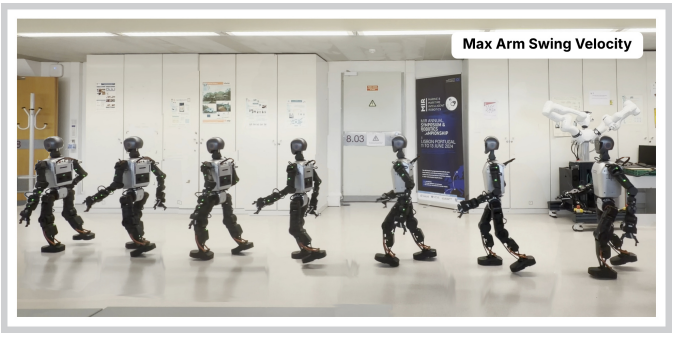


Fig. 13. Deployment of the Reference Motion Policy on the Real T1 at the Maximum Forward Command, with the Wide, Articulated Arm Swing Validated in Simulation (Video).

The policy transferred faithfully onto the real robot (Fig. 13), with the wide swing at maximum speed virtually indistinguishable from simulation, the elbow following it within 1.9–3.3° RMSE, a reduced swing at medium speed, and the arms returning smoothly to rest during deceleration, with only small corrective adjustments.

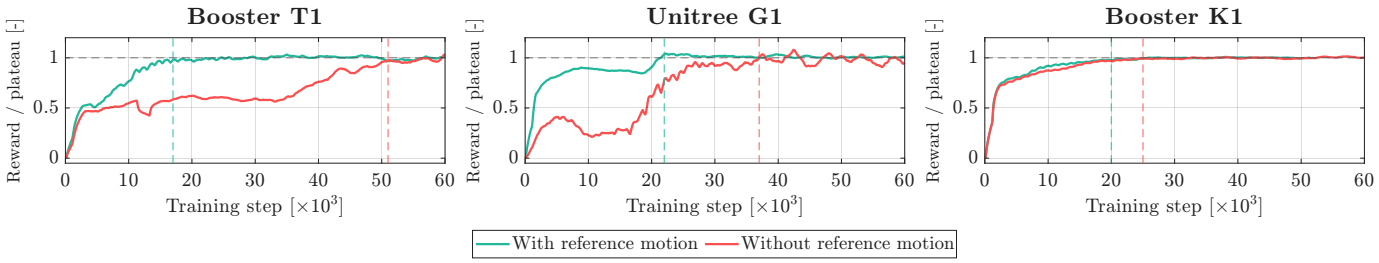


Fig. 10. Normalized Training Reward of the Walking Policy With (Green) and Without (Red) the Reference Motion, for the T1, G1 and K1. Each curve is rescaled from its initial value to its plateau, so the curves compare how fast training stabilizes. Dashed lines mark convergence.

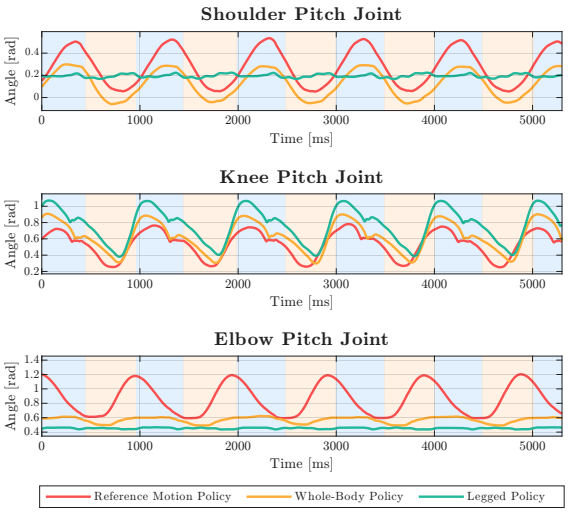


Fig. 14. Right Shoulder Pitch, Elbow Pitch and Knee Pitch of the Legged, Whole-Body and Reference Motion T1 Policies in Real Deployment. Shading marks right and left foot forward.

The robot does not simply walk faster or slower, it walks differently. At the maximum command, the real robot swings its heavy arms through this wide range without any motor limitation, and the movement is noticeably more coordinated and life-like than in the earlier whole-body deployment. Compared with the whole-body policy (Fig. 14, Table III), the shoulder and elbow follow strong, smooth and coordinated waves, while the legs adopt a less crouched posture, with mean knee flexion falling from 37.8° to 29.3°.

Ankle power fell by 53% and lateral sway decreased, with ankle roll range falling from 27.9° to 15.8°.

TABLE III: Leg Load per Stride: Whole-Body vs Reference Motion Policy on the Real Robot.

Metric	Whole-Body	Ref. Motion	Change
Leg power [W]	20.1	19.0	−6%
Ankle power [W]	13.6	6.4	−53%
Ankle roll range [°]	27.9	15.8	−43%
Mean knee flexion [°]	37.8	29.3	−22%
Peak knee flexion [°]	64.9	49.5	−24%
L–R similarity, legs [%]	74.7	80.8	+8%
L–R similarity, arms [%]	46.8	95.3	+104%

In the legged policy, holding the arms against gravity made the shoulder and elbow correct continuously, which appears as small, uneven oscillations, and freeing them removed most of this irregularity. The knee pitch shows that more upper-body motion is consistently accompanied by less knee flexion across the objectives. The largest reduction occurs from the legged to the whole-body policy, where the robot no longer counteracts the momentum of its walk with rigidly held arms, and a further, smaller one follows with the reference motion. Left-right similarity of the arm joints rose from 46.8% to 95.3%, with the elbow rising from 51.2% to 98.1%, and that of the legs from 74.7% to 80.8%. The wider arm movement makes this policy less efficient overall than the whole-body one, but it places the lowest load on the legs, which receive the most strenuous forces. Where preserving the legs is the priority, as they are the joints most prone to wear, it may therefore be the preferable option. The contribution of this objective is thus not simply a larger arm swing, but a movement that is more structured across joints and between sides, with velocity-dependent upper-body motion and a lower load on the legs, which becomes the walking foundation for the stair objective.

E. Perceptive Stair Traversal

The climbing policy approaches with the regular walking gait and, at each step, lifts the lead leg with pronounced hip and knee flexion, places it on the tread, and brings the trailing leg up beside it (Fig. 5).

This is a much more strenuous motion than walking, using considerably larger joint ranges and torques.



Fig. 17. Ascent Validation in Isaac Sim across All Studied Robots (Video).

Six robots climbing the same flight of 15 cm steps converge to the same basic strategy (Fig. 17). The lead leg flexes strongly at the hip and knee at the start of each step, and the trailing leg follows with a smaller excursion (Fig. 15). The smaller T1 and K1 reach about twice the hip flexion of the taller A3 and GR3, and the K1 needs more than twice as long per step, taking extra repositioning steps on each tread. The narrow deviation bands show that each robot repeats its own pattern consistently from one step to the next.

Over full 12-step flights at three tread depths, in both MuJoCo and Isaac Sim, the T1 ascends in virtually every episode up to 25 cm steps, well beyond the 18 cm regulated range, after which success collapses within 2–3 cm (Fig. 16). The limit is nearly the same across tread depths and simulators, so it reflects the robot's physical reach.

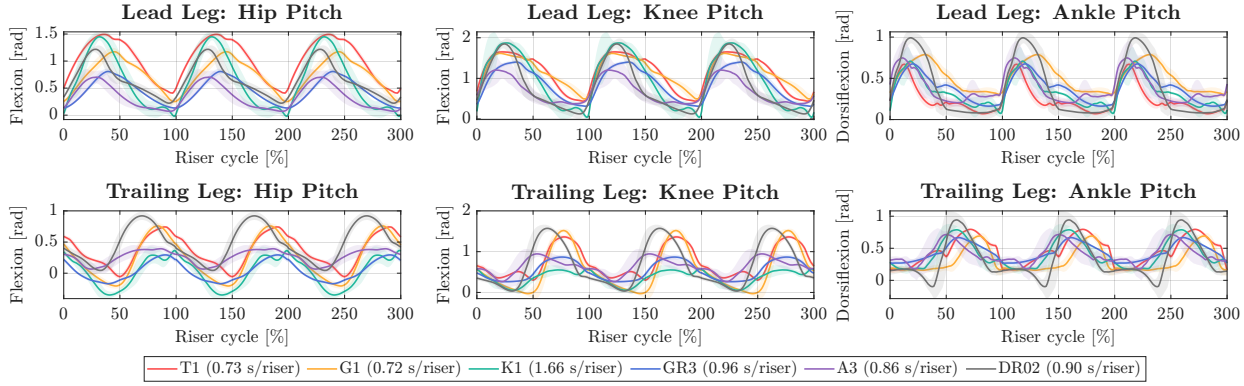


Fig. 15. Leg Pitch Joints of Six Robots Climbing the Same Flight of Twelve 15 cm Steps with 0.4 m Treads at a Commanded 0.5 m/s. Each cycle runs from one lead-leg lift-off to the next, so it climbs exactly one step. Lines show the mean and bands ± 1 standard deviation across step cycles.

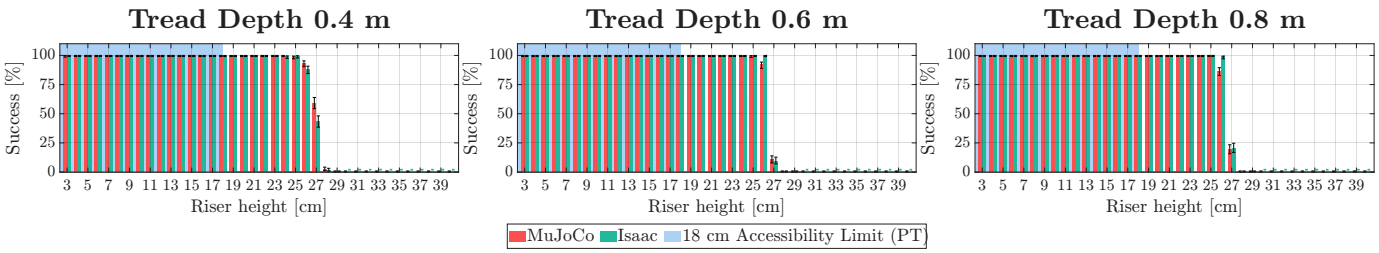


Fig. 16. T1 Stair Ascent Success in MuJoCo and Isaac Sim across Step Heights and Tread Depths. Each condition contains 400 randomized-start episodes on a 12-step staircase. Error bars show 95% Wilson intervals.

TABLE IV: Ascent Success at High Steps and Measured Stepping Behavior. Lift is the peak sole height above the tread left, relative to leg length L. Swing RMS is the deviation of each swing from the mean swing, lower meaning more repeatable.

Robot	Success [%] 24 cm	28 cm	32 cm	Tread time [s]	Lead lift [% L]	Swing RMS [% L]
T1	98	0	0	1.17	39.8	4.2
K1	4	0	0	2.14	54.3	15.9
G1	99.5	47	0	1.24	33.3	5.7
GR3	100	92	0	1.18	26.4	7.4
A3	100	100	41	1.42	28.0	5.7
DR02	100	96	24	1.02	43.7	10.9
Kangaroo	67	20	0	1.58	38.1	6.4

Across the other robots, the reachable heights follow leg length (Table IV), from about 21 cm for the K1 to above 28 cm for the A3 and DR02, and all show near-perfect success within the regulated range. The taller GR3, DR02 and A3 also decay more gradually past their limit, over 6–10 cm, whereas the T1 drops within about 2 cm. The Kangaroo's parallel leg linkage gives it a reach similar to the T1's despite its size. The K1 keeps a very high success rate up to 21 cm, climbing entire flights of steps taller than 23% of its height. The arms take part in the climb without any reward forcing it, with their angular momentum opposing that of the body for about 60% of the ascent on the T1 and G1. Relative to leg length, the K1 and DR02 lift their feet the highest, and the T1 has the most repeatable swing of the group.

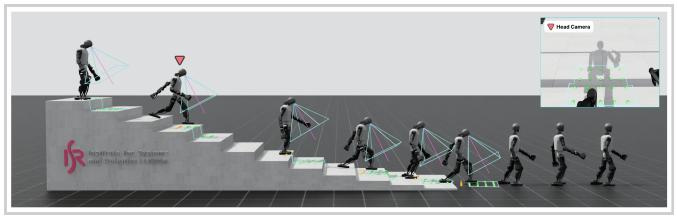


Fig. 18. Stair Descent of the Booster T1 Guided by its Head Camera. Green heightmap samples are observed in the current frame, yellow ones are read from memory, and limb occlusions are filtered out. (Video)

Adding descent did not degrade ascent, and a single policy performs both, distinguishing them from the heightmap. Descent needs different placements and timings, with the focus on never speeding through the steps, so the robot must be able to halt its momentum at every drop. Going down, it uses long horizontal strides with the full hip range to plant firmly, and the arms counterbalance to stop the forward momentum at each drop (Fig. 18). At these heights the robot stands on a single foot while fully spread, so the arms even out its weight before the next step. The figure also shows the perception at work, with the head turned down to see as much of the heightmap as possible, rows out of view read from memory, and limb occlusions filtered out.

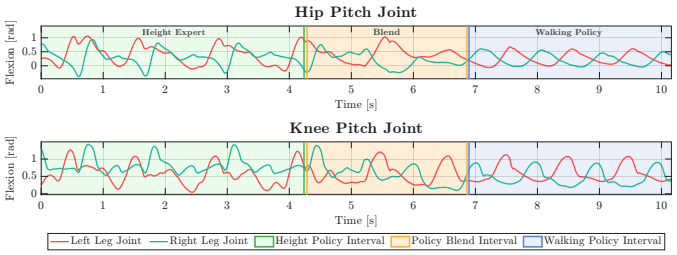
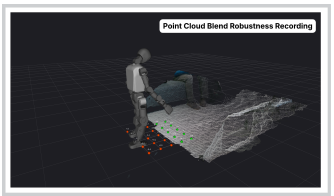


Fig. 19. T1 Hip and Knee Pitch across the Hand-off from Stair Descent to Walking. Shading marks the policy in control: height expert, blend and walking policy. The uneven stair gait gives way to regular, alternating walking cycles of about 1 s.

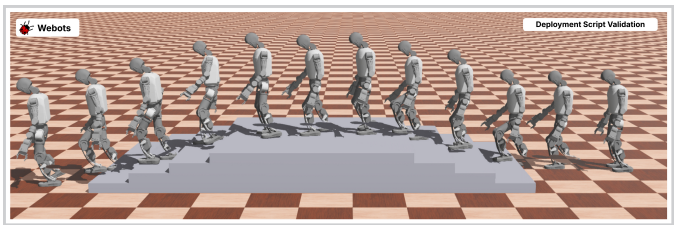
Descent reaches the same robustness as ascent except on the shallowest 0.4 m treads, barely longer than the robot's feet, where momentum accumulates over steps on 14–18 cm drops until a placement is missed.



(a) Simulated Head Camera in MuJoCo.



(b) Real Point Clouds Replayed in Simulation.



(c) Deployment Script in Webots through the Booster SDK.

Fig. 20. Validation of the Camera-Based Stair Pipeline. Heightmap samples are green when observed in the latest depth frame and orange when read from the map's memory.

When the heightmap no longer detects steps, control returns to the walking policy through a blend of about 2.8 s, and the joints settle into the regular 1 s cycles of the reference gait within another 1.5 s, without the robot stopping or moving erratically (Fig. 19).

The DR02, with the longest reach, additionally learned the alternating stride over two treads where geometry allows, on 0.40–0.50 m treads within a set height interval and at its maximum leg reach, falling back to step-to climbing elsewhere. This lunge needs the longest reach of the group, as the shorter, bulkier legs of the T1

leave no margin to touch down two treads ahead with the required precision.

F. Camera-Based Perception

The heightmap imperfections cost little in performance: with exact heights, ascent progress rose by only 8%, episodes without a fall by 7%, and descent progress not at all. With the heightmap reconstructed from a simulated head camera, with its depth noise, dropouts and latency, the robot climbed with the same controlled stepping (Fig. 20a). In Webots, through the SDK, the deployment script climbed four steps, crossed a landing and descended four more in one continuous run, with limb occlusions removed, remembered rows covering the area under the feet, and the blend handing back to the walker after the last step (Fig. 20). On 58 s of point clouds recorded by the real head camera and replayed in simulation, the depth feed delivered every frame at 30 Hz with a longest gap of 46 ms, well below the drops randomized in training, every far-row cell received depth returns in every frame, and the blend held the walking policy on flat floor for all 2,880 control steps without a single false detection.

V. Conclusions and Future Work

This work pursued natural and robust locomotion for the Booster T1, a humanoid whose long, heavy arms make both harder to obtain.

We first developed legged and whole-body controllers for this asymmetric morphology. The legged policy transferred directly to the real robot with a wider, more fluid and more consistent gait than the official controller. Extending it to the whole body showed that the arms are an asset rather than a burden, with 41% lower mechanical power and 29.5% lower torque-rate RMS than the official controller over a sustained trial.

Second, the energy efficiency study showed that the restrained swing is the economical one for the T1, and that across five humanoids the economical swing follows each robot's morphology rather than a human-like pattern. Efficiency alone therefore does not lead every robot to a human-like gait, which motivated specifying the desired gait explicitly.

Third, retargeting adapted to non-human proportions and a velocity-indexed reference library supplied what rewards could not. The deployed gait is more pronounced at high speed and restrained at low speed, more symmetric and less crouched, with lower ankle effort, and on the T1 it was learned in 65% fewer training steps.

Finally, a perceptive climbing policy built from human sub-movements ascends steps of up to 25 cm in validation, well beyond the regulated 18 cm, and descends with similar robustness except on the shallowest treads. It blends with the walking policy without either expert degenerating, and the reachable heights of the other robots follow their leg length.

The stair traversal was validated in simulation, including the camera-to-heightmap pipeline, the deployment script in Webots and real camera data replayed in simulation, but not yet on the real robot. Camera noise and latency, and the larger joint ranges and torques of climbing, demand a far more exact transfer than walking. Real deployment of the stair and blending system, with extensive robustness trials, is therefore the most immediate step. Further directions include step-level operator control on stairs, mapping the staircase at a distance so the robot need not look down, a dedicated policy for small adjustments before a step, and extending the same approach to manipulation on top of the stabilized locomotion.

Declaration

To compress some of the text to meet the maximum set page count, AI tools were utilized. More specifically, Claude Opus by Anthropic.

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